

HIGH-DIMENSIONAL ASYMPTOTICS

Mostly [Mei21]:

1. MODEL PROBLEMS

Classical inference problems deal with constant dimensionality d and sample size n tending to ∞ , so that the design matrix only grows taller. Here we consider the proportional regime, where d scales with n . The design matrix grows in both directions. We consider Bayesian inference problems with distributions that resemble statistical physics models.

Problems: $\mathbb{Z}_2$ synchronization, spiked GOE, lasso regression, ridge regression.

$\mathbb{Z}_2$ sync: we observe $Y = \lambda \frac{\theta\theta^T}{n} + W$, where $W \sim GOE(n)$ and $\theta \sim Unif(\{\pm 1\}^n)$ (Bayesian), and we want to recover θ . We will let θ denote the true (random) parameter, and σ a generic element of the discrete cube. (Why $\frac{1}{n}$? Because the leading eigenvalue of W is of constant order, as we will show rigorously in the RMT section. For a rough understanding, note that $\langle v, Wv \rangle$ is $N(0, \frac{1}{n})$ for a single $v \in S^{n-1}$, and taking a sup over $v \in S^{n-1}$ corresponds to a sup of 2^n iid gaussians, which incurs an additional factor of $\sqrt{\log 2^n} \propto \sqrt{n}$. On the other hand, the leading eigenvalue of the projector $\theta\theta^T$ is $|\theta|^2 = n$, so we scale it down by n .)

Density of $GOE(n) \propto \exp(-\frac{1}{4}n|W|_F^2)$. Likelihood $L(\sigma|Y) \propto \exp(-\frac{1}{4}n|Y - \lambda \frac{\sigma\sigma^T}{n}|_F^2)$. Posterior on $\theta \propto \exp(-\frac{1}{4}n|Y - \lambda \frac{\sigma\sigma^T}{n}|_F^2)\nu(d\sigma)$.

MLE of θ : the negative log likelihood is proportional to $|Y - \lambda \frac{\sigma\sigma^T}{n}|_F^2$. Recall that $\langle A, B \rangle_F = \text{tr} A^T B = \text{tr} A B^T$ (the dot product between the matrices). Expanding, we obtain $|Y|_F^2 + \frac{\lambda^2}{n^2}|\sigma|^4 - \frac{2\lambda}{n}\langle Y, \sigma\sigma^T \rangle_F$, which reduces to $|Y|_F^2 + \frac{\lambda^2}{n^2} - \frac{2\lambda}{n}\langle Y\sigma, \sigma \rangle$. Thus $\hat{\theta}_{ML} = \text{argmax}_{\sigma} \langle \sigma, Y\sigma \rangle$.

Spectral estimator of θ : relax the MLE to the sphere of radius $\sqrt{n}$. This yields $\hat{\theta}_{sp} = \text{argmax}_{\sigma \in S^{n-1}(\sqrt{n})} \langle \sigma, Y\sigma \rangle = \sqrt{n}v_{max}(Y)$. We analyze it below.

Bayes estimator of $\theta\theta^T$: from the above, the posterior is $\propto \exp(\frac{1}{2}\lambda\langle \sigma, Y\sigma \rangle)$, or equivalently $\propto \exp(\frac{\lambda}{2}\langle \sigma, W\sigma \rangle + \frac{\lambda^2}{2n}\langle \sigma, \theta \rangle^2)$. The exponent is a quadratic form in σ , so the posterior is a (random) Gibbs measure, specifically a perturbed SK measure. We could let $H(\sigma) = -\frac{1}{2}\langle \sigma, W\sigma \rangle - \frac{\lambda}{2n}\langle \sigma, \theta \rangle^2$ and set $\beta = \lambda$. Note that the MLE derived above is a ground state. The Bayes estimator becomes $\langle \sigma\sigma^T \rangle$.

Spiked GOE: we observe $Y = \lambda uu^T + W$, where $W \sim GOE(n)$ and $u \sim Unif(S^{n-1})$, and we want to recover u .

Lasso: consider the linear model $y = Ax_0 + w$, where the design matrix $A \in \mathbb{R}^{n \times d}$ has iid $N(0, \frac{1}{n})$ entries, the entries of x_0 are iid P_0 (Bayesian), and $w \sim N(0, \sigma^2 I_n)$. The lasso estimator of x_0 is ℓ_1 -regularized least squares: $\hat{x} = \operatorname{argmin}_x \frac{1}{2n} |y - Ax|_2^2 + \frac{\lambda}{n} |x|_1$. The lasso estimator is typically used when we believe x_0 to be sparse. (Notation is a bit unusual, since typically X is the design matrix and β_0 is the parameter.)

Ridge: again a linear model, but notation switches to $Y = X\beta_0 + w$, where X has $N(0, 1)$ entries, $\beta_0 \sim N(0, \frac{1}{d} I_d)$ (Bayesian), and $w \in N(0, \sigma^2 I_n)$. The estimator of β_0 is ℓ_2 -regularized least squares: $\hat{\beta} = \operatorname{argmin}_\beta \frac{1}{2n} |Y - X\beta|^2 + \frac{\lambda d}{2n} |\beta|^2$. (Why the d ? Because Y has entries of order 1, whereas β_0 has entries of order $\frac{1}{\sqrt{d}}$.) This has explicit solution $\hat{\beta} = (X^T X + \lambda d I_d)^{-1} X^T Y$.

2. FREE ENERGY TRICK

A common trick in statistical physics is to recover the magnetization from the free energy. As an example, for the Ising model with Hamiltonian $H_h(\sigma) = -\sum_{i \sim j} J_{ij} \sigma_i \sigma_j + h \sum_i \sigma_i$, we have $\partial_h F(\beta, h) = -\frac{1}{\beta} \frac{\partial_h Z}{Z} = \langle \sum_i \sigma_i \rangle_{\beta, h}$.

There's nothing special about this choice of Hamiltonian. In general, if we want to compute the expectation of an observable $M(\sigma)$ with respect to a Gibbs measure defined by an (unperturbed) Hamiltonian $H(\sigma)$, we can define the perturbed Hamiltonian $H_\lambda(\sigma) = H(\sigma) + \lambda M(\sigma)$. By the above, $\langle M(\sigma) \rangle_{\beta, \lambda} = \partial_\lambda F(\beta, \lambda)$. Setting $\lambda = 0$, we deduce $\langle M(\sigma) \rangle_\beta = \partial_\lambda F(\beta, 0)$. This is the free energy trick.

In light of the free energy trick, we see that computing the free energy (or equivalently, the partition function) is fundamental.

Example: $\mathbb{Z}_2$ sync.

What is the expected overlap $\frac{1}{n^2} \langle \hat{\theta}_{sp}, \theta \rangle^2$ between $\hat{\theta}_{sp}$ and θ ? (Inside $E[\cdot]$, we can treat θ as a constant.) We consider the observable $M(\sigma) = \frac{1}{n} \langle \sigma, \theta \rangle^2$, which we will evaluate at $\hat{\theta}_{sp}$. Since $\hat{\theta}_{sp}$ is defined as a sup, if we let $H_\lambda(\sigma) = -\frac{1}{2} \langle \sigma, Y \sigma \rangle$, then the Gibbs measure on configurations $\sigma \in S^{n-1}$ with density $\propto \exp(-\beta(H_\lambda(\sigma) + hM(\sigma)))$ has ground state $\hat{\theta}_{sp}$. (A configuration is a single vector on the sphere.) By the free energy trick, $\langle M \rangle_{\beta, \lambda} = \partial_h F(\beta, \lambda, 0)$. Taking the low temperature limit $\beta \rightarrow \infty$, the Gibbs measure concentrates on its ground state $\hat{\theta}_{sp}$, so that $\langle M \rangle_{\beta, \lambda} \rightarrow M(\hat{\theta}_{sp})$. Thus the asymptotic expected overlap is

$$\lim_{n \rightarrow \infty} E\left[\frac{1}{n} M(\hat{\theta}_{sp})\right] = \lim_{\beta \rightarrow \infty} \lim_{n \rightarrow \infty} \frac{1}{n} \partial_h F(\beta, \lambda, 0) = \lim_{\beta \rightarrow \infty} \partial_h f(\beta, \lambda, 0) \quad (1)$$

where f is the free energy density. If we computed f , we would discover the BBP phase transition: for $\lambda \leq 1$, the limit equals 0, while for $\lambda > 1$ the limit equals $1 - \frac{1}{\lambda^2}$. Thus $\hat{\theta}_{sp}$ has no overlap with θ if the SNR is too low, and has overlap tending to 1 as $\lambda \rightarrow \infty$. This makes sense, because if λ is large, then the projector $\frac{\lambda}{n} \theta \theta^T$ dwarfs W , and the

leading eigenvector is approximately θ . (In physics terms, the BBP phase transition is a phase transition in the observable M in the spin system.)

What is the expected quadratic risk of the estimator Θ_{Bayes} of $\theta\theta^T$? (For convenience we drop the subscript.) Write $|\Theta - \theta\theta^T|_F^2 = 1 + |\Theta|_F^2 - 2\langle\theta, \Theta\theta\rangle$. We seek $a_* := \lim \frac{1}{n^2} E\langle\theta, \Theta\theta\rangle$ and $c_* := \lim \frac{1}{n^2} E|\Theta|_F^2$.

We start with a_* . Since $\Theta = \sum \sigma \sigma^T P(\sigma|Y)$, we have $\langle\theta, \Theta\theta\rangle = \sum \langle\sigma, \theta\rangle^2 P(\sigma|Y)$. This is the average of $\langle\sigma, \theta\rangle^2$ with respect to the posterior $P(\sigma|Y)$, which is a Gibbs measure. We define the observable $M(\sigma) = \frac{1}{2n} \langle\sigma, \theta\rangle^2$, the perturbed Hamiltonian $H_\lambda(\sigma) = -\frac{1}{2} \langle\sigma, W\sigma\rangle + \lambda M(\sigma)$, and we note that the posterior $P(\sigma|Y)$ is the Gibbs measure with $\beta = \lambda$. We have $\frac{1}{n^2} \langle\theta, \Theta\theta\rangle = -\frac{2}{n} \langle M \rangle_{\lambda, \lambda}$. By the free energy trick, $-\frac{2}{n} \langle M \rangle_{\beta, \lambda} = -2\partial_\lambda f(\beta, \lambda)|_{\beta=\lambda}$. If we could compute f , we would obtain a_* . c_* can be calculated similarly.

3. REPLICA METHOD

We saw that computing the free energy is fundamental. However, the logarithm is difficult to handle.

Replica trick: $E[\log Z] = \lim_{k \rightarrow 0} \frac{1}{k} \log EZ^k$. (Heuristic: $E[\log Z] \approx E[\frac{Z^k - 1}{k}] = \frac{EZ^k - 1}{k} \approx \frac{1}{k} \log EZ^k$.)

The name “replica” comes from the fact that Z^k is the partition function of k independent copies of the system. Importantly, each of these replicas experience the same quenched randomness, because the expectation $E[\cdot]$ over the quenched randomness is outside Z^k . In the case of the SK model, this means that the coupling constants are identical for each replica. The weirdest part of the heuristic replica trick is the limit $k \rightarrow 0$. In some cases it has been made rigorous.

Example: $\mathbb{Z}_2$ sync.

Let θ be drawn uniformly at random from the discrete cube, and let $P_{\beta, \lambda}$ be the $\mathbb{Z}_2$ sync Gibbs measure at inverse temperature β (so not necessarily the posterior). We want to characterize the joint distribution of (σ, θ) . We compute

$$m_*(\beta, \lambda) = \lim_n E\left[\frac{1}{n} \sum_{i=1}^n \langle \psi(\sigma_i, \theta_i) \rangle_{\beta, \lambda}\right] \quad (2)$$

where the outer expectation is over θ, W . As an application, this allows us to compute the expected quadratic risk of the estimator $\langle\sigma\rangle_{\beta, \lambda}$ of θ .

By the free energy trick, it suffices to compute the limiting free energy density (or free entropy density) of the Gibbs measure with perturbed Hamiltonian $H_{\lambda, h}(\sigma) = -\langle\sigma, Y\sigma\rangle - h \sum_{i=1}^n \psi(\sigma_i, \theta_i)$. (Missing factors in the notes?)

This gets messy. Assuming the replica trick works, we write the free entropy as $\lim_n \frac{1}{n} E[\log Z_n] = \lim_n \lim_k \frac{1}{nk} \log EZ_n^k$. We first compute $S(k, \beta, \lambda, h) = \lim_n \frac{1}{n} \log EZ_n^k$.

Then we compute the free entropy density $\phi(\beta, \lambda, h) = \lim_k \frac{1}{k} S$. Finally we compute $m_*(\beta, \lambda) = \frac{1}{\beta} \partial_h \phi(\beta, \lambda, 0)$ (the $\frac{1}{\beta}$ coming from entropy versus energy).

To compute S , note that Z_n^k is the integral of $\exp(-\beta \sum_{a=1}^k H(\sigma^a))$, where the replicas $\sigma^1, \dots, \sigma^k$ are iid $\sim \text{Unif}(\{\pm 1\}^n)$. Importantly, the coupling constants in $H(\sigma^a)$ are the same for all a . The expectation of the quadratic gaussian term $\langle \sigma^a, W \sigma^a \rangle$ in $H(\sigma^a)$ can be computed via the mgf. The remaining terms are nonrandom. To handle terms like $\exp(\frac{1}{n} \beta^2 \langle \sigma^a, \sigma^b \rangle^2)$, we use the delta function identity trick: $\int_{\mathbb{R}} \delta(x - q) dq = 1$ for any $x \in \mathbb{R}$. In other words, we replace $\langle \sigma^a, \sigma^b \rangle$ with q_{ab} and add integration with respect to $\delta(\langle \sigma^a, \sigma^b \rangle - q_{ab}) dq_{ab}$. To handle terms like $\exp(\frac{1}{n} \beta \lambda \langle \sigma^a, \sigma^a \rangle^2)$, we introduce the variables q_{0a} . (In optimization, we often do something similar: replace annoying quantities with new variables, and modify the objective to enforce the constraint.) This adds integration variables with q_{ab} , where $0 \leq a, b \leq k$. (For $1 \leq a, b \leq k$, (a, b) corresponds to a pair of replicas, and q_{0a} corresponds to their overlap. Not clear how to interpret q_{0a} ?)

By an application of a saddle point approximation, we obtain

$$S = \sup_Q \inf_{\Lambda} U(Q, \Lambda) \quad (3)$$

where U is a complicated function of $Q, \Lambda \in \mathbb{R}^{(k+1) \times (k+1)}$ that is quadratic in Q . To solve the FOC of this optimization problem, we need to make an ansatz. It turns out that U has permutation symmetry in the variables q_{ab} with $1 \leq a, b \leq k$ associated to the k replicas. This leads us to make the replica symmetric ansatz Q , add...

(Isn't the lower-right $k \times k$ submatrix of Q supposed to be the overlap matrix between the replicas? The overlap matrix was mentioned in passing in the large deviations section. Why didn't we compute Q apriori?)

Example: one can also apply the replica trick to the lasso risk.

What is RSB? A more complicated ansatz?, add...

4. GAUSSIAN COMPARISON

Motivation: lasso risk. If we want to compute $m_* = \lim_{d \rightarrow \infty} E[\frac{1}{d} \sum_{i=1}^d \psi(\hat{x}_i, x_{0,i})]$ for some loss function ψ , we can introduce the perturbed Hamiltonian $H_{\lambda,h}(x) = \frac{1}{2n} |y - Ax|_2^2 + \frac{\lambda}{n} |x|_1 + h \frac{1}{d} \sum_{i=1}^d \psi(x_i, x_{0,i})$. (No minus sign.) The lasso estimator is the ground state of the Gibbs measure at $\beta = \infty$. By the free energy trick, $\langle \frac{1}{d} \sum_i \psi(x_i, x_{0,i}) \rangle_{\beta, \lambda, h} = \partial_h F(\beta, \lambda, 0)$. As $\beta \rightarrow \infty$, the free energy converges to the energy of the ground state. (Justification: $-\frac{1}{\beta} \log Z = -\frac{1}{\beta} \log \sum_x e^{-\beta H(x)} \approx -\frac{1}{\beta} \log e^{-\beta H(\hat{x})} = H(\hat{x})$.) Sending $\beta \rightarrow \infty$, we obtain $\frac{1}{d} \sum_{i=1}^d \psi(\hat{x}_i, x_{0,i}) = \partial_h \big|_{h=0} \min_x H_{\lambda,h}(x)$.

Let us rewrite the limiting free energy $\min_x H_{\lambda,h}(x)$ slightly. We use a variational trick: sometimes it's useful to expand a simple expression into a complicated variational form. For example, the Legendre transform of $\frac{1}{2} |v|_2^2$ is itself, so that $\frac{1}{2} |u|_2^2 = \max_v \langle u, v \rangle - \frac{1}{2} |v|_2^2$.

We write

$$\min_x \frac{1}{2} \|y - Ax\|_2^2 + \lambda \|x\|_1 = \min_x \max_v \langle y - Ax, v \rangle - \frac{1}{2} \|v\|^2 + \lambda \|x\|_1 \quad (4)$$

Because the objective is a linear function of the gaussian RM A , we are left with an extremum of a Gaussian process indexed by (x, v) . We need some qualitative comparison theorems.

(The first GP that comes to my mind is BM. But BM is a very complicated and rough process. The simplest GP is $X_t = \langle t, g \rangle$, whose realizations are smooth linear functions of t .)

Sudakov-Fernique: if $(X_t), (Y_t)$ are GPs on $T \subseteq \mathbb{R}^d$ with common mean and such that $E[(X_t - X_s)^2] \leq E[(Y_t - Y_s)^2]$, then $E[\sup X_t] \leq E[\sup Y_t]$.

Slepian: if we also have common variance $EX_t^2 = EY_t^2$, then in fact we have stochastic domination: $\sup X_t \stackrel{sd}{\leq} \sup Y_t$.

Sudakov-Fernique is intuitive. The more independent a GP is, the more likely it is to attain higher values. The covariance condition implies that the values of Y are less correlated than the values of X . (The natural metric $d(t, s) = (E|X_t - X_s|^2)^{1/2}$ on T induces an interesting geometry.)

(Proof by interpolation when X, Y are n -dimensional gaussian random vectors. We define the approximation $F_\beta(x) = \beta^{-1} \log(\sum_i e^{\beta x_i})$, which as $\beta \rightarrow \infty$ tends to $\max_i x_i$. (Think back to the free energy at low temperature.) If we show $E[F_\beta(X)] \leq E[F_\beta(Y)]$, we will be done. Our interpolation from X to Y is given by $Z(\theta) = X \cos \theta + Y \sin \theta$, which is again gaussian. We write

$$F(Y) - F(X) = \int_0^{\pi/2} \partial_\theta F(Z(\theta)) d\theta \quad (5)$$

$$= \int_0^{\pi/2} \langle \nabla F(Z(\theta)), Z'(\theta) \rangle d\theta \quad (6)$$

Note that $Z'(\theta) = -X \sin \theta + Y \cos \theta$. Thus the expectation of the integrand equals

$$\sum_i -\sin \theta E[\partial_i F(Z(\theta)) X_i] + \sum_i \cos \theta E[\partial_i F(Z(\theta)) Y_i] \quad (7)$$

Recall multivariate gaussian IBP: $E[X_i f(X)] = \sum_j \text{Cov}[X_i X_j] E[\partial_j f(X)]$. This yields

$$\cos \theta \sin \theta \sum_{i,j} (E[Y_i Y_j] - E[X_i X_j]) E[\partial_{ij} F(Z(\theta))] \quad (8)$$

We compute $\partial_i F(x) = p_i := \frac{e^{\beta x_i}}{\sum_i e^{\beta x_i}}$, so that $\nabla F(x)$ is a Gibbs-like pmf. Next $\partial_{ij} F(x) = \beta p_i \mathbf{1}_{i=j} - \beta p_i p_j$. This equals $\beta(p_i - p_i^2)$ when $i = j$ and $-\beta p_i p_j$ otherwise. We split the sum into $i = j$ and $i \neq j$, move the $-\beta p_i^2$ terms over to the $i \neq j$ sum, introduce a sum on j in

the first sum, and swap indices i, j in the first sum:

$$= \beta \sum_i (E[Y_i^2] - E[X_i^2])E[p_i] + \beta \sum_{i,j} (E[Y_i Y_j] - E[X_i X_j])E[p_i p_j] \quad (9)$$

$$= \beta \sum_{i,j} (E[Y_i^2] - E[X_i^2])E[p_i p_j] + \beta \sum_{i,j} (E[Y_i Y_j] - E[X_i X_j])E[p_i p_j] \quad (10)$$

$$= \frac{\beta}{2} \sum_{i,j} (E[(Y_i - Y_j)^2] - E[(X_i - X_j)^2])E[p_i p_j] \quad (11)$$

which is clearly nonnegative. (Missing $E[\cdot]$ around p_i 's in notes?)

We apply Sudakov-Fernique by comparing a complicated X to a simple Y , or vice versa.

Example: largest singular value of gaussian RM, cheat by using the CGMT comparison, add...

Gordon's inequality is a generalization to the minmax of a GP, add...

CGMT: let $S_u \subseteq \mathbb{R}^m$, $S_v \subseteq \mathbb{R}^n$ be compact. Let $G \in \mathbb{R}^{m \times n}$, $g \in \mathbb{R}^n$, $h \in \mathbb{R}^m$ have iid standard gaussian entries. To understand $\Phi(G) = \min_u \max_v \langle u, Gv \rangle + \psi(u, v)$, compare it to $\phi(g, h) = \min_u \max_v |u|_2 \langle g, v \rangle + |v|_2 \langle h, u \rangle + \psi(u, v)$. We have $P(\Phi(G) \geq \tau) \leq 2P(\phi(g, h) \geq \tau)$ (typo in notes?), and if S_u, S_v are convex, then $P(\Phi(G) \leq \tau) \leq 2P(\phi(g, h) \leq \tau)$ as well.

(Proof by Gordon, but you need to make a clever choice of X , add...)

Finally, we apply CGMT to the lasso risk. Plugging in $y = Ax_0 + w$, $A = \frac{1}{\sqrt{n}}G$, and writing $u = x_0 - x$, our GP becomes

$$\Phi(G) = \frac{1}{\sqrt{n}} \langle v, Gu \rangle + \psi(u, v) \quad (12)$$

where $\psi(u, v) = \langle v, w \rangle - \frac{1}{2}|v|_2^2 + \lambda|x_0 - u|_1$. (Here we condition on the noise w ?) By CGMT, we compare to

$$\frac{1}{\sqrt{n}}|u|_2 \langle g, v \rangle + \frac{1}{\sqrt{n}}|v|_2 \langle h, u \rangle + \psi(u, v) \quad (13)$$

We start with the inner max over v . The objective is approximately quadratic of the form $\langle a, v \rangle + b|v|_2 - \frac{1}{2}|v|_2^2$, which apparently has maximum $\frac{1}{2}(|a|_2 + b)_+^2$. We set $a = \frac{1}{\sqrt{n}}|u|_2 g + w$ and $b = \frac{1}{\sqrt{n}} \langle h, u \rangle$. By concentration, $|g|_2^2 \approx n$ and $|w|_2^2 \approx \sigma^2 n$, so that $|a|_2 \approx (\frac{|u|_2^2}{n} + \sigma^2)^{1/2} \sqrt{n}$. Our problem becomes

$$n \cdot \min_u \frac{1}{2} \left(\left(\frac{|u|_2^2}{n} + \sigma^2 \right)^{1/2} + \frac{1}{n} \langle h, u \rangle \right)_+^2 + \frac{\lambda}{n} |x_0 - u|_1 \quad (14)$$

The biggest obstacle to optimizing in u is the ReLU term. So we call $|u|_2^2$, $\langle h, u \rangle$ new variables and replace the objective with the Lagrangian. The minimizer in u can then apparently be computed as

$$\operatorname{argmin}_u \lambda|x_0 - u|_1 + \frac{\mu}{2}|u|_2^2 - \nu \langle h, u \rangle = \eta \left(x_0 + \frac{\nu}{\mu} h; \frac{\lambda}{\mu} \right) - x_0 \quad (15)$$

where $\eta(x; s) = \text{sign}(x)(|x| - s)_+$ is the soft thresholding function. When this is plugged in, you can write down the FOCs to solve for the dual/slack variables. Then you can evaluate the asymptotics of the risk L .

Other applications of CGMT listed in Xinmeng's talk, add...

5. RMT

Motivation: understand the empirical distribution $\hat{\mu}_A = \frac{1}{n} \sum_i \delta_{\lambda_i(A)}$ of the eigenvalues of an $n \times n$ random matrix A as $n \rightarrow \infty$.

The most common transform in probability theory is the characteristic function. However, there is a more useful transform to study eigenvalue distributions.

Given a measure μ on $\mathbb{R}$, its Stieltjes transform is the function $m : \mathbb{C} \setminus \text{supp} \mu \rightarrow \mathbb{C}$ given by the real integral $m(z) = \int_{\mathbb{R}} \frac{1}{x-z} \mu(dx)$.

Note that the Stieltjes transform of $\hat{\mu}_A$ is

$$m(z) = \frac{1}{n} \sum_i \frac{1}{\lambda_i(A) - z} = \frac{1}{n} \text{tr}(A - zI)^{-1} \quad (16)$$

The matrix $(A - zI)^{-1}$ is known as the resolvent of A at z .

It will be useful to know $\Im m(z)$, which can be computed as

$$\Im m(z) = \int_{\mathbb{R}} \frac{\Im(\overline{x-z})}{|x-z|^2} d\mu(x) = \int_{\mathbb{R}} \frac{\eta}{(x-\lambda)^2 + \eta^2} d\mu(x) \quad (17)$$

As a first observation, note $\eta > 0$ implies $\Im m(z) > 0$, so that m sends the upper half-plane into itself. Next, if we let p_η denote the density of the Cauchy(0, η) distribution, then $\frac{1}{\pi} \Im m(z) = (p_\eta * \mu)(\lambda)$. (The equation in the notes doesn't seem correct?) If the scale parameter η is very small, then we are essentially convolving μ with the delta measure δ_0 , and we expect $\frac{1}{\pi} \Im m(z)$ to approximately be the density of μ at λ (if such a density exists). Thus we can recover μ through integration:

$$\lim_{\eta \rightarrow 0} \int_a^b \frac{1}{\pi} \Im m(z) d\lambda = \mu([a, b]) \quad (18)$$

This is known as Stieltjes inversion, the analogue of Levy's inversion theorem for the characteristic function. The Stieltjes transform preserves all the information in μ .

We also have an analogue of Levy's continuity theorem. The weak convergence $\mu_n \xrightarrow{w} \mu$ is equivalent to the pointwise convergence $m_n(z) \rightarrow m(z)$ for all z in the upper halfplane. (This holds true even when μ_n are random probability measures, in which case weak convergence reads $\mu_n f \xrightarrow{as} \mu f$ for all nice f .)

We now analyze the empirical eigenvalue distribution of $GOE(n)$. We have heuristically argued that its eigenvalues are of constant order. Now we show that the empirical distribution converges to the semicircle law μ_{sc} with density $\frac{1}{2\pi}\sqrt{4-x^2}\mathbf{1}_{x\in[-2,2]}$, supported on $[-2, 2]$.

One can show that the Stieltjes transform of μ_{sc} equals $m = \frac{\sqrt{z^2-4}-z}{2}$. (Change variables from x to $2\cos\theta$, rewrite as a rational function of e^{it} , and convert to a contour integral about the unit circle.) This satisfies the quadratic equation $m^2 + zm + 1 = 0$.

We want to show that in the limit, the Stieltjes transform of μ_n obeys this same equation. To compute the Stieltjes transform of μ_n , we use the “leave one out” trick, which relates m_n to m_{n-1} . Suppose $W \sim GOE(n)$. We compute each diagonal entry of $(W - zI_n)^{-1}$. Swap row i and row 1 of $W - zI_n$, and then swap column i and column 1. (Why is this allowed?) The $(1, 1)$ entry of the modified resolvent will equal the (i, i) entry of the original resolvent. Write $W - zI_n = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, where $A = W_{ii} - z$, where $B^T = C = \omega_i$ has iid $N(0, \frac{1}{n})$ entries, and where $D = W^{(i)} - zI_{n-1}$ is the lower $(n-1) \times (n-1)$ submatrix. The block matrix inversion formula shows that the $(1, 1)$ entry of $(W - zI_n)^{-1}$ is

$$(A - BD^{-1}C)^{-1} = \frac{1}{W_{ii} - z - \omega_i^T(W^{(i)} - zI_{n-1})^{-1}\omega_i} \quad (19)$$

Here ω_i and $W^{(i)}$ are independent, so assuming that the quadratic form concentrates, we have

$$\omega_i^T(W^{(i)} - zI_{n-1})^{-1}\omega_i \approx E[\omega_i^T(W^{(i)} - zI_{n-1})^{-1}\omega_i | W^{(i)}] \quad (20)$$

$$= \frac{1}{n} \text{tr}(W^{(i)} - zI_{n-1})^{-1} \quad (21)$$

Note that $W^{(i)} \sim \frac{\sqrt{n-1}}{\sqrt{n}}GOE(n-1)$, so this quantity is approximately $m_{n-1}(z)$. Thus $(W - zI_n)_{ii}^{-1} \approx \frac{1}{W_{ii} - z - m_{n-1}}$. Averaging over i and ignoring $W_{ii} = O(\frac{1}{\sqrt{n}})$, we obtain $m_n(z) \approx \frac{1}{-z - m_{n-1}(z)}$, which implies $m_n^2 + zm_n + 1 \approx 0$.

(A very similar method can be used to derive MP law via Stieltjes; see C-D for details...)

Example: double descent in ridge regression.

The bias-variance tradeoff for the MSE of an estimator states that $MSE = \text{bias}^2 + \text{variance}$. As we increase model complexity, the bias typically decreases but the variance increases. (The expectation is over all possible datasets we are presented with. We do better at estimating the parameter on average, but our estimator depends more strongly on the particular dataset we fit it to.) Thus the MSE should look convex. But in highly overparametrized models, the MSE starts decreasing again, and this is called double descent. (In the case of polynomial regression, it's intuitive.)

For ridge regression, it's not clear to me if “model complexity” is the right word, add... We want to compute the MSE as a function of γ , and at the end we set $\lambda = 0$.

If we condition on β_0 and X , then the conditional MSE can be simplified as

$$= |E[\hat{\beta}|\beta_0, X] - \beta_0|^2 + \text{Var}[\hat{\beta}|\beta_0, X] \quad (22)$$

$$= |(X^T X + \lambda d I)^{-1} X^T X - I| \beta_0|^2 + E[|(X^T X + \lambda d I)^{-1} X^T Y|^2 | \beta_0, X] \quad (23)$$

$$= |\lambda d (X^T X + \lambda d I)^{-1} \beta_0|^2 + E[|(X^T X + \lambda d I)^{-1} X^T Y|^2 | \beta_0, X] \quad (24)$$

To obtain the MSE, we take an expectation over β_0, X . We consider the bias first. Using the trace trick, this equals

$$E[|\lambda d (X^T X + \lambda d I)^{-1} \beta_0|^2] \quad (25)$$

$$= (\lambda d)^2 E[\text{tr}(X^T X + \lambda d I)^{-1} \beta_0 \beta_0^T (X^T X + \lambda d I)^{-1}] \quad (26)$$

$$= \lambda^2 d E[\text{tr}(X^T X + \lambda d I)^{-2}] \quad (27)$$

$$= \lambda^2 \frac{1}{d} E[\text{tr}(\frac{1}{d} X^T X + \lambda I)^{-2}] \quad (28)$$

In terms of the spectrum of $\frac{1}{d} X^T X$ (known as a Wishart matrix, a multidimensional analogue of χ_d^2), this equals

$$= \lambda^2 E\left[\int \frac{1}{(s + \lambda)^2} d\hat{\mu}_d(s)\right] \quad (29)$$

It turns out that $\hat{\mu}_d$ converges weakly to a distribution called the Marchenko-Pastur law, which allows this to be computed directly. We can also go via the Stieltjes transform of $\hat{\mu}_d$. If $m(z) = \text{tr}(\frac{1}{d} X^T X - zI)^{-1}$, then $m'(z) = -\text{tr}(\frac{1}{d} X^T X - zI)^{-2}$, so that the bias equals $-\lambda^2 m'(-\lambda)$. The variance can be handled similarly, but an auxiliary argument must be added to the Stieltjes transform.

List of important RM ensembles, CDE, nonasymptotic theory from Ramon's lecture, add...

6. FREE PROBABILITY

(Cite Couillet-Debbah, Tao, Biane talk...):

In classical probability, if A, B are independent real-valued rv's, then the distribution of $A + B$ is the convolution of the distribution of A with the distribution of B .

What about for rm's? For rm's, the natural distribution on $\mathbb{R}$ is the esd. Is there an easy expression for the esd of $A + B$ in terms of the esd's of A and B ? (We don't expect anything nice. If A, B hermitian, then they're scaling along ONBs. The only nice case is when those two ONBs align, i.e. A, B simultaneously diagonalizable and the evals add. But in general, a nonlinear operation.)

If A, B are "freely independent", yes. The result is the free convolution $\mu_A \boxplus \mu_B$ whose R-transform obeys $R = R_A + R_B$.

How to formulate free independence? Two classical rv's are independent if $E[f(A)g(B)] = E[f(A)]E[g(B)]$ for all nice f, g . (Think polynomials.)

Let $\tau := \frac{1}{n} E \text{tr}$. This will serve as the natural expectation when working with non-commutative rm's. (Why this τ ? It fits the natural distribution, the esd. It yields moments of the expected esd: if $E[\frac{1}{n} \sum_i \delta_{\lambda_i}]$ is the expected esd of A , then $\tau(A^k) = E[\frac{1}{n} \sum_i \lambda_i^k]$ is k -th moment of expected esd.) We say two rm's A, B are freely independent if $\tau(f_1(A)g_1(B) \cdots f_k(A)g_k(B))$ splits as a product for all polynomials f_i, g_i . This is not the same as classical independence; $\tau(ABAB) \neq \tau(A^2)\tau(B^2)$. If A, B freely independent, you get formulas for all joint moments in terms of individual moments. We can define independence of ≥ 3 rm's in a similar way.

Another viewpoint: for a single rv A , the moments $E[A^k]$ determine the distribution. For a rm A , the free moments $\tau(A^k)$ determine the expected esd. Similarly, joint free moments determine joint expected esd's (?).

A weaker notion is asymptotic freeness of sequences A_n, B_n . This allows us to characterize lsd of $A_n + B_n$ in terms of lsd's of A_n, B_n , just as above. Asymptotic freeness defn: the same as above, except (1) we require limit distributions for each of A_n, B_n , and (2) only require the difference to go to zero as dimension $n \rightarrow \infty$. Generalizes to ≥ 3 rm's of course.

Natural connection to method of moments: if asympt freeness holds, then the limiting joint distribution is a product distribution. This is determined by its mixed moments, so compute all of these to characterize lsd. (Ex: if we want lsd of $A_n + B_n$, compute $\tau((A + B)^k)$ for all k using $\tau(A^l), \tau(B^l)$, and figure out what distribution has these moments. In free CLT, the answer is semicircle law.)

Some classically indep sequences are asympt free; but not generally true.

Ex of classically indep: for each n , consider $A_{n,i}$ = independent Wigner matrices.

Another ex of classically indep: for each n , consider $A_{n,i} = U_i D_i U_i^*$ for $i \in [m]$, where U_i are iid sampled according to Haar measure on $U(n)$ and where D_i is a deterministic real diagonal. The A_i 's have same evals but different evects. The sets of evects are completely independent. Then $\{A_{n,1}, \dots, A_{n,m}\}$ are asympt free. (According to C-D, this is the intuitive defn of asympt freeness: espaces are maximally uncorrelated; thus two deterministic matrices can't be free...) (Biane makes geometric analogy to iid vectors on S^n ...)

It can be hard to check asympt freeness. Certain sufficient conditions, like in Thm 4.5 of C-D. See the corollary for details, but includes operations involving gaussian matrices, unitary matrices, conjugating a hermitian by a gaussian/unitary...

To avoid method of moments, can bundle them together into a generating func. (Recall R transform is essentially the holo part of the inverse of Stieltjes transform, i.e. $R = m^{-1}(-z) - \frac{1}{z}$. For big z , by inspecting Laurent series, $m(z) \sim -1/z$.) Why is R transform additive for free rm's?

Formal derivation from Tao (changing notation): let X, Y be free. (Why does this calculation use $\tau = \frac{1}{n} \text{tr}$, not $\tau = \frac{1}{n} E \text{tr}$? Both are valid τ 's for the non-commutative

probability space, but the former yields esd while latter yields expected esd? Since Stieltjes transform is defined in terms of τ , this choice affects defn of Stieltjes. Typically functionals of esd concentrate around expected esd?...

$$s = s_X(z) \quad (= \tau((X - zI)^{-1}))$$

Invert this map: $z = z_X(s)$, so that $s_X(z_X(s)) = s$. This z_X will almost be the R transform.

$$\tau((X - z_X(s)I)^{-1}) = s$$

$$(X - z_X(s)I)^{-1} = s(I - E_X) \text{ for a rm } E_X \text{ with trace } 0$$

$$\text{Solve for } X: X = z_X(s)I + s^{-1}(I - E_X)^{-1}$$

$$\text{Sum the same identity for } Y: X + Y = (z_X(s) + z_Y(s))I + s^{-1}((I - E_X)^{-1} + (I - E_Y)^{-1})$$

$$\text{By factoring we get } X + Y = (z_X(s) + z_Y(s) + s^{-1})I + s^{-1}(1 - E_X)^{-1}(1 - E_X E_Y)(1 - E_Y)^{-1}$$

$$\text{To identify what } z_{X+Y}(s) \text{ is, we rearrange: } ((X + Y) - (z_X(s) + z_Y(s) + s^{-1})I)^{-1} = s(1 - E_Y)(1 - E_X E_Y)^{-1}(1 - E_X)$$

Take τ : by freeness, the RHS becomes s . So $z_{X+Y}(s) = z_X(s) + z_Y(s) + s^{-1}$. This is not quite additive, but $R(s) = z(-s) - s^{-1}$ is.

R transform is the analog of the log characteristic function in classical probability (which is additive for classically indep rv's). R transform can be expanded as power series (b/c inverting Laurent series and killing $1/z$ term?). By analogy its coefficients are "free cumulants".

App: rederive special case of Thm 3.13 in C-D...

Can formulate general non-commutative probability spaces, beyond those consisting of rm's. Inspiration: instead of thinking about probability space $(\Omega, \mathcal{F}, \mathbb{P})$, consider algebra of functions on this space (rv's) with expectation operator. Abstraction: $(\mathcal{A}, \tau)$ for unital $\mathbb{C}$ -algebra $\mathcal{A}$ and linear functional τ s.t. $\tau(1) = 1$. For rm's, we work on the algebra $(\mathcal{A}, \tau) = (L^\infty(\Omega) \otimes M_n(\mathbb{R}), \frac{1}{n} E \text{tr})$ (or $\frac{1}{n} \text{tr}$).

Limitations: free probability only tells you about $n \rightarrow \infty$, not rates/finite dimension results; asymp freeness only holds for certain sequences of rm's...

7. UNIVERSALITY

Lindeberg swapping is one way to prove the CLT. Given $U_1, \dots, U_n$ iid from a distribution with mean zero and variance 1, we want to show weak convergence to $N(0, 1)$, i.e. $E[g(\frac{1}{\sqrt{n}} \sum_i U_i)] \rightarrow E[g(N(0, 1))]$ for any nice $g \in C(\mathbb{R})$. Equivalently, we want to show that $E[g(\frac{1}{\sqrt{n}} \sum_i U_i)] - E[g(\frac{1}{\sqrt{n}} \sum_i V_i)] \rightarrow 0$, where V_i iid $N(0, 1)$. Lindeberg swapping estimates this difference by interpolating between the vectors U and V . We let W_i be

the vector $(U_1, \dots, U_i, V_{i+1}, \dots, V_n)$ obtained by swapping out the first i entries of V with those of U . If we define $f : \mathbb{R}^n \rightarrow \mathbb{R}$ by $f(x) = g \circ (x \mapsto \frac{1}{\sqrt{n}} \sum_i x_i)$, then we can write

$$E[f(U) - f(V)] = \sum_i E[f(W_{i+1}) - f(W_i)] \quad (30)$$

Note that W_{i+1} and W_i only differ in their i -th entry. By Taylor expanding each of $f(W_{i+1}), f(W_i)$ around $W_i^0 := (U_1, \dots, U_{i-1}, 0, V_{i+1}, \dots, V_n)$ (not around W_i !), and by matching the moments of U_i, V_i , we obtain $E[f(W_{i+1}) - f(W_i)] \leq \frac{1}{6} \|\partial_i^3 f\|_\infty (E[V_i^3] - E[U_i^3])$. This is $O(n^{-3/2})$, just enough to conquer the n terms.

Chatterjee generalized the Lindeberg principle beyond the context of weak convergence. The theorem provides conditions on f for $|E[f(U)] - E[f(V)]|$ to be small when U and V are vectors of iid random variables. When this holds, f is said to exhibit universality, in that the distribution of its inputs don't asymptotically affect its output.

Example: universality of free entropy of SK model. add...

8. BP AND AMP

[Mei21]: these notes present BP, MP, and AMP in terms of distributions. Specifically, we are given a Gibbs measure that defines a factor graph, and the goal is to compute the marginal of x_i . The derivation of BP in this setting is very similar to below. Imagine the factor graph is a tree. Consider the marginal of i in $a \rightarrow i$, denoted $\mu_{a \rightarrow i}(x_i)$. This is a sum over all other variables $x_k, k \neq i$. We factor out the terms involving i and regroup the remaining terms in this sum to obtain $\mu_{a \rightarrow i}(x_i) \propto \sum_{x_{\partial a \setminus i}} \psi_a(x_{\partial a}) \prod_{j \in \partial a \setminus i} \mu_{j \rightarrow a}(x_j)$. (Note i is not a node in $a \rightarrow i$ but x_i still appears in the factor ψ_a ...) To compute $\mu_{i \rightarrow a}(x_i)$, you regroup the terms based on factors $b \in \partial i \setminus a$ to obtain $\mu_{i \rightarrow a}(x_i) \propto \psi_i(x_i) \prod_{b \in \partial i \setminus a} \mu_{b \rightarrow i}(x_i)$.

If you simplify these equations by assuming the messages are gaussian, then you can simply pass the mean and variance. This is MP. Finally, AMP is a tricky mean field approximation which passes a single (m, v) pair from each node. (In statistical physics, AMP can be used to directly estimate the magnetization.) (The version below can be thought of as a low temperature limit of distributional AMP, where we compute the ground state.)

[MEK12]: we want to recover x from a noisy observation $y = Ax + w$. We utilize a Bayesian framework. Assume the prior P_0 on x is a product distribution, perhaps each entry is nonzero with small probability ε . Call this class $\mathcal{F}_\varepsilon$. Let the noise distribution be gaussian $N(0, \sigma^2 I_n)$. The $m \times n$ sensing matrix A is fixed, and we are in the proportional regime $m/n \rightarrow \delta$. If $m = n$ and $A = I_n$, then this reduces to a denoising problem, and the denoising perspective will be useful below.

One particular algorithm to recover x is the lasso, which encourages sparsity. Consider the 1d case with $A = 1$. To recover x from $y = x + w$, we solve $\operatorname{argmin}_z \frac{1}{2}(y - z)^2 + \lambda|z|$.

This has solution $\hat{x} = \eta(y; \lambda)$, where the threshold θ equals the regularization λ . We still need to choose λ , and intuitively we should scale $\lambda \propto \sigma$.

What is the intuition for the soft thresholding function η ? Well, the typical size of w is σ . If we observe $y \gg \sigma$, then it's more likely that w is positive, so we subtract something $\propto +\sigma$. If we observe $y \ll -\sigma$, then w must have contributed $\propto -\sigma$, so we add something. If y is in $[-\sigma, \sigma]$, we set it to zero. (Is this intuition OK?)

One can determine the minimax choice of threshold in terms of MSE. It turns out that soft thresholding does very well at denoising gaussian noise.

To solve lasso and analyze the lasso estimator in higher dimensions, it's not so clear. We can't just solve the FOC as in 1d. We could try traditional methods (see [Tib15]): to optimize lasso, you can try subgradient descent, which would lead to unthresholded updates like $x^{t+1} \approx x^t + A^T r^t - \lambda v$ for some subgradient v of $x \mapsto |x|_1$ at x^t . An algo with a fast convergence guarantee is proximal GD, or ISTA. Recall for a convex differentiable function, GD with step size α can be rewritten as

$$x^{t+1} = \operatorname{argmin}_x f(x^t) + \nabla f(x^t)^T (x - x^t) + \frac{1}{2\alpha} |x - x^t|_2^2 \quad (31)$$

That is, we approximate f by its linearization about x^t plus a quadratic term. As a sanity check, let's complete the square. The minimum of the quadratic occurs at $-\alpha \nabla f(x^t)$, so the objective function should be (up to an additive constant) $\frac{1}{2\alpha} |x - (x^t - \alpha \nabla f(x^t))|_2^2$. This confirms that $x^{t+1} = x^t - \alpha \nabla f(x^t)$, the GD iteration. Now suppose $f = g + h$ where g is convex differentiable but h is merely convex. We can replace g with its quadratic approximation and try to optimize the resulting objective. The PGD algorithm sets

$$x^{t+1} = \operatorname{argmin}_x \frac{1}{2\alpha} |x - (x^t - \alpha \nabla g(x^t))|_2^2 + h(x) \quad (32)$$

In the case of lasso, this separates into a bunch of 1d lasso problems. We find x^{t+1} by component-wise soft thresholding: $x^{t+1} = \eta(x^t - \alpha_t A^T (Ax^t - y); \lambda \alpha_t)$. If we define the residual $r^t = y - Ax^t$, we get the ISTA updates $x^{t+1} = \eta(x^t + \alpha_t r^t; \lambda \alpha_t)$.

However, the iterates of ISTA are not easy to analyze. Instead we use an iterative algorithm to solve the optimization problem. When approximated, it yields MP and AMP. The distribution of the iterates x^t of AMP can be characterized.

Goal: minimize a cost function $C(x) = \sum_{a \in F} C_a(x_{\partial a}) + \sum_{i \in V} C_i(x_i)$, where F are factors and V are variables. These are arranged in a factor graph. (The lasso objective is $\frac{1}{2} \|y - Ax\|_2^2 + \lambda \|x\|_1$, or $\sum_{a \in [m]} \frac{1}{2} (y_a - A_a^T x)^2 + \sum_{i \in [n]} \lambda |x_i|$. This has a complete bipartite factor graph, with one a for each measurement, and one variable x_i for each entry of x .)

Min-cost algo: unlike [Mei21] presentation, we don't pass distributions, but instead pass functions. We want to determine $J(x_i)$, the min value of the objective fixing the value of x_i . If we can determine $J(x_i)$, then we can determine x_i^* as $\operatorname{argmin} J(x_i)$. Suppose the factor graph is a tree. Suppose variable i is a leaf, so that it is attached to a single factor a . The factor graph can be decomposed as the disjoint union of all branches meeting

at a . Let the sub-factor graph corresponding to $j \in \partial a$ be denoted $j \rightarrow a$. These are all nodes reachable from j without passing through a . Then grouping the terms of the objective function, we have the equality (1): $J(x_i) = \min_{x_{\partial a \setminus i}} \{C_i(x_i) + C_a(x_{\partial a}) + \sum_{j \in \partial a \setminus i} J_{j \rightarrow a}(x_j)\}$. We have reduced the computation of $J(x_i)$ to the computation of $J_{j \rightarrow a}(x_j)$ in the smaller factor graph $j \rightarrow a$. We would like to recursively apply (1) to compute each $J_{j \rightarrow a}(x_j)$. Unfortunately, it need not be the case that j is a leaf in $j \rightarrow a$. We split the factor graph $j \rightarrow a$ into $b \rightarrow j$ for each $b \in \partial j \setminus a$. Here $b \rightarrow j$ denotes the sub-factor graph consisting of all nodes reachable from b without passing through j . (But it still includes the term $C_b(x_{\partial b})$?...) We have (2): $J_{j \rightarrow a}(x_j) = C_j(x_j) + \sum_{b \in \partial j \setminus a} J_{b \rightarrow j}(x_j)$. In $b \rightarrow j$, j is a leaf, so we can apply (1) to compute $J_{b \rightarrow j}(x_j)$. This yields $J(x_i)$ in a single pass through the graph. The above procedure can be thought of as message passing. $J_{b \rightarrow j}$ is passed from each $b \in \partial j \setminus a$ to j ; using this information, j passes $J_{j \rightarrow a}$ to a for all $j \in \partial a \setminus i$; finally, a passes $J(x_i)$ (later denoted $J_{a \rightarrow i}$) to i .

For a general graph, we can still perform such a procedure, but it won't yield the truth in a single pass. The iterations are:

$$J_{a \rightarrow i}^t(x_i) = \min_{x_{\partial a \setminus i}} \{C_a(x_{\partial a}) + \sum_{j \in \partial a \setminus i} J_{j \rightarrow a}^t(x_j)\} \quad (33)$$

$$J_{i \rightarrow a}^{t+1}(x_i) = C_i(x_i) + \sum_{b \in \partial i \setminus a} J_{b \rightarrow i}^t(x_i) \quad (34)$$

($C_i(x_i)$ doesn't appear in first equation because i is not included in $a \rightarrow i$?...) The first equation is the analogue of (1) above; it computes $J(x_i)$ in the sub-factor graph $a \rightarrow i$. The second equation is the analogue of (2) above; it splits $i \rightarrow a$ into $b \rightarrow i$ for all $b \in \partial i \setminus a$. It's important to note that these iterations are non-backtracking. (Note $a \rightarrow i$ is computed using current time messages, while $i \rightarrow a$ is computed using previous time messages.)

To simplify min-sum for lasso, we approximate each message function by a quadratic function. Assume the columns of A are mean zero and have ℓ_2 norm 1, so that $A_{ai} \approx O(\frac{1}{\sqrt{m}})$. Let's start with $J_{a \rightarrow i}^t$. It depends on x_i through the product $A_{ai}x_i$. Write

$$J_{a \rightarrow i}^t(x_i) = -\alpha_{a \rightarrow i}^t(A_{ai}x_i) + \frac{1}{2}\beta_{a \rightarrow i}^t(A_{ai}x_i)^2 + O((A_{ai}x_i)^3) \quad (35)$$

Plugging into $J_{i \rightarrow a}^{t+1}(x_i)$, since we have incoming messages from all $b \neq a$, we have

$$J_{i \rightarrow a}^{t+1}(x_i) = \lambda|x_i| - \left(\sum_{b \neq a} A_{bi}\alpha_{b \rightarrow i}^t\right)x_i + \frac{1}{2}\left(\sum_{b \neq a} A_{bi}^2\beta_{b \rightarrow i}^t\right)x_i^2 + o(1) \quad (36)$$

To approximate $J_{i \rightarrow a}^{t+1}$ as a quadratic, we need to find its minimum (notice the $\lambda|x_i|$ term). This is absolute value + quadratic, so its minimum is given by soft thresholding. By matching coefficients with the expression above, we get

$$J_{i \rightarrow a}^{t+1}(x_i) = \frac{1}{2\gamma_{i \rightarrow a}^{t+1}}(x_i - x_{i \rightarrow a}^{t+1})^2 + O(\dots) \quad (37)$$

where $x_{i \rightarrow a}^{t+1} = \eta(a_1; a_2)$ and $\gamma_{i \rightarrow a}^{t+1} = \eta'(a_1; a_2)$ for some a_1, a_2 that are complicated functions of α^t, β^t . We can also determine α^t, β^t as functions of γ^t, x^t . As time t grows large, we expect $x_i^t \approx x_i^*$, so it's good that we see the soft thresholding value as in 1d lasso. In fact, the threshold a_2 is λ times some scaling constant.

We simplify once more to obtain AMP. In MP, for each variable i , we pass an estimate $x_{i \rightarrow a}^*$ of x_i to each factor a . It would be nicer if we only kept track of one such estimate x_i^* . Given thresholds (θ_t) and reaction terms (b_t) , the AMP iterates are the vectors

$$x^{t+1} = \eta(x^t + A^T r^t; \theta_t) \quad (38)$$

$$r^t = (y - Ax^t) + b_t r^{t-1} \quad (39)$$

We expect the estimate x^{t+1} to involve soft thresholding, as in 1d lasso and MP. The residual r^t is a bit strange, in that it includes a momentum term $b_t r^{t-1}$.

Important intuition: although our original problem $y = Ax + w$ is not strictly speaking a denoising problem, there is a neat denoising interpretation of AMP. Consider the unthresholded estimate of x at time t given by $x^t + A^T r^t$. This is x + some “effective noise”. Going from step t to $t + 1$, we see that AMP applies the soft thresholding function η to $x^t + A^T r^t$ to remove this noise. Thus AMP is iterative denoising of successive estimates of x .

Why is this related to the lasso estimator? Suppose $\theta_t = \theta$ and $b_t = b$. Consider a fixed point (x^*, r^*) . Then $(1 - b)r^* = y - Ax^*$ and $x^* = \eta(x^* + A^T r^*; \theta)$. From the definition of η , the second equation reads $x^* = (x^* + A^T r^*) - \theta v$, where $v_i = \text{sgn}(x_i)$. Rearranging, we obtain $\theta v - A^T r^* = 0$, or equivalently $(1 - b)\theta v + A^T(Ax - y) = 0$. This is the FOC of lasso when $\lambda = (1 - b)\theta$. So if we can choose $(\theta_t) \rightarrow \theta$, $(b_t) \rightarrow b$ such that x^t converges, we reach the lasso solution.

Derivation of AMP: we need to simplify iterations of α, β, γ, x , but we need to be careful for which terms we drop i, a dependence.

First, we consolidate α, β into the residual r . It's natural to define $r_{a \rightarrow i}^t = \frac{\alpha_{a \rightarrow i}^t}{\beta_{a \rightarrow i}^t}$, since this approximately leaves the a -th component of the residual at iterate t . The threshold a_2 defining $x_{i \rightarrow a}^{t+1} = \eta(a_1; a_2)$ can be approximated by LLN as θ_t , independent of i, a . We obtain $r_{a \rightarrow i}^t = y_a - \sum_{j \neq i} A_{aj} x_{j \rightarrow a}^t$ and $x_{i \rightarrow a}^{t+1} = \eta(\sum_{b \neq a} A_{bi} r_{b \rightarrow i}^t; \theta_t)$.

How do we approximate r, x dynamics? We can't immediately assume all messages sent from a given starting point are all equal. I.e. in our recurrence, we can't immediately replace $x_{i \rightarrow a}^t$ with a single x_i^t for all a , and we can't replace $r_{a \rightarrow i}^t$ with a single r_a^t for all i . This would yield the incorrect iterations $x = \eta(A^T r; \theta)$ and $r = y - Ax$. (This is missing $x + A^T r$ in x iteration, missing Onsager term...)

The non-backtracking is important. Write $r_{a \rightarrow i}^t = r_a^t + \delta r_{a \rightarrow i}^t$ and $x_{i \rightarrow a}^t = x_i^t + \delta x_{i \rightarrow a}^t$, where the δ terms are $O(\frac{1}{\sqrt{n}})$. We keep these small terms in the recursion to see how they contribute. We compute recursions for the mean field term and the perturbation.

Match terms on both sides of the r equation to find $\delta r_{a \rightarrow i}^t = A_{ai}x_i^t$. (This makes sense; we originally exclude the term $A_{ai}x_{i \rightarrow a}^t$ from the r recurrence, and since $x_{i \rightarrow a}^t = x_i^t + O(\frac{1}{\sqrt{n}})$, using x_i^t only incurs error $O(\frac{1}{n})$.)

For the x equation, first Taylor expand. Then match terms for the x equation to obtain $\delta x_{i \rightarrow a}^{t+1} = -\eta'(\dots; \theta_t)A_{ai}r_a^t$, the impact of including the $A_{ai}r_{a \rightarrow i}^t$ term inside η .

Now that the perturbations $\delta r, \delta x$ are in terms of r, x , we can eliminate them from our mean field r, x recursions. This yields

$$x_i^{t+1} = \eta\left(\sum_b A_{bi}r_b^t + \sum_b A_{bi}^2x_i^t; \theta_t\right) \quad (40)$$

$$r_a^t = (y_a - \sum_j A_{aj}x_j^t) + \sum_j A_{aj}^2\eta'(\dots; \theta_{t-1})r_a^{t-1} \quad (41)$$

In the x recursion, $\sum_b A_{bi}^2 \rightarrow 1$, so upon approximation it becomes the i -th component of $x^{t+1} = \eta(A^T r^t + x^t; \theta_t)$. In the r recursion, since $A_{aj} = O(\frac{1}{\sqrt{m}})$, we can replace the correction term with an average over j . Since $\eta'(x; \theta) = \text{sgn}(|\eta(x; \theta)|)$, we can replace $\eta'(\dots; \theta_{t-1})$ with $\text{sgn}(|x_j^t|)$, which when averaged over j yields $\frac{1}{m}|x^t|_0$. We obtain the Onsager term $+b_t r^{t-1}$ with $b_t = \frac{1}{m}|x^t|_0$, so that $r^t = (y - A^T x^t) + b_t r^{t-1}$, as desired.

State evolution: as [Mei21] puts it, to analyze AMP dynamics (which involves high-d vectors), it suffices to study certain low-d dynamics. Specifically, it suffices to study the “effective noise” τ_t of the estimate at time t . τ_t^2 is approximately the variance of the error in our unthresholded estimate at time t : $(x^t + A^T r^t) - x \approx N(0, \tau_t^2 I)$. (This is not the same as the variance of $x^t - x$ at time t ; that yields $\tilde{\tau}_t^2 = \lim_n \frac{1}{n}|x^t - x|^2$.) The precise definition is the recursion $\tau_{t+1}^2 = F(\tau_t^2, \theta_t)$, where the function F is

$$F(\tau^2, \theta) = \sigma^2 + \frac{1}{\delta} E[(\eta(X_0 + \tau Z; \theta) - X_0)^2] \quad (42)$$

for $(X_0, Z) \sim P_0 \times N(0, 1)$, and with initial condition $\tau_0^2 = \sigma^2 + \frac{1}{\delta} E X_0^2$.

Sanity check: at time 0, our iterate is just $y = Ax + w$, which has covariance matrix $\sigma^2 I + E[AA^T]EX_0^2$. Since the columns of A are iid $N(0, \frac{1}{m}I)$, this is approximately $(\sigma^2 + \frac{1}{\delta} EX_0^2)I$. The recurrence from t to $t+1$ can be derived from AMP iterations. It's convenient to utilize $\tilde{\tau}_t$. Let's consider $(x^t + A^T r^t) - x$. Approximate $r^t \approx y - Ax^t = w - A(x^t - x)$ and plug this in. We obtain $(x^t + A^T r^t) - x = A^T w + (I - A^T A)(x^t - x)$. Treat the terms as independent. Note $A^T w$ has covariance matrix $\sigma^2 E[A^T A] \approx \sigma^2 I$. The entries of $I - A^T A$ are approximately iid $N(0, 1/m)$ (?). Thus $(I - A^T A)(x^t - x)$ is approximately $N(0, \frac{1}{\delta} \tilde{\tau}_t^2)$. This implies that $\tau_t^2 = \sigma^2 + \frac{1}{\delta} \tilde{\tau}_t^2$, and also that $x^t + A^T r^t \approx X_0 + \tau_t Z$ for $(X_0, Z) \sim P_0 \times N(0, 1)$. Plugging the latter into the x recurrence, $x^{t+1} - x \approx \eta(X_0 + \tau_t Z; \theta_t) - X_0$, and taking the variance yields $\tilde{\tau}_{t+1}^2 = E[(\eta(X_0 + \tau_t Z; \theta_t) - X_0)^2]$. We now eliminate $\tilde{\tau}_t^2$ from the equation for τ_t^2 above. This implies $\tau_{t+1}^2 = \sigma^2 + \frac{1}{\delta} E[(\eta(X_0 + \tau_t Z; \theta_t) - X_0)^2]$, as desired.

Thm 6.1/6.3 show that (τ_t) determine the joint distribution of AMP iterates and the true x , as well as the joint distribution of the lasso estimator (AMP fixed point) and the

true x . From this one can derive the lasso risk as $\tilde{\tau}_*^2$, or in terms of τ_*^2 , the expectation $E[(\eta(X_0 + \tau_* Z; \theta_*) - X_0)^2]$.

(Q: Why is it more natural to consider τ_t instead of $\tilde{\tau}_t$?...)

General Q: which other optimization problems amenable to AMP?

General Q: isn't compressed sensing about ℓ_0 constraint, not ℓ_1 regularization?

Other things: TAP, Kac-Rice? add...

[FMM21]: the TAP free energy (a function of the magnetization vector, i.e. the vector of average spins) is a good approximation to the SK free energy, includes Onsager term, a pure state corresponds to a local minimum of the TAP free energy, it is hypothesized that the SK measure is a convex combination of pure states, there are many local minima of TAP free energy, replica symmetry = well approximated by a single pure state, this paper shows that for $\mathbb{Z}_2$ sync replica symmetry holds in that the global minimizer of the TAP free energy is a good approximation of the posterior mean, GD is a natural algorithm to find the minimum but currently only AMP methods are known to work...

9. MISCELLANEOUS

[Pri24]:

Computing the free energy density of the SK model with external magnetic field h was a big open problem. Sherrington-Kirkpatrick invented the replica method to heuristically compute this quantity. However, their method only works in the replica symmetric region. Aizenman-Lebowitz-Ruelle proved the SK formula in the replica symmetric region with $h = 0$ and $\beta < 1$. Talagrand extended the result to $h > 0$ and $\beta < 1/2$. Then Talagrand proved the heuristic Parisi formula for the free energy, which holds for all $\beta, h > 0$, including the region of replica symmetry breaking.

Equivalence of ensembles: we have been describing statistical mechanics systems by a Gibbs measure $\propto \exp(-\beta H(\sigma))$ over microstates σ . This is known as the canonical ensemble at a given inverse temperature β . A more fundamental description of a system is the microcanonical ensemble. In this formulation, the energy of the system is fixed (not the temperature), and all microstates are given an equal probability. It turns out that these two ensembles are equivalent. Although we don't explicitly fix the energy of the macrocanonical ensemble, it happens to concentrate around its expectation, the internal energy U . This is the origin of the particular density $\exp(-\beta H(\sigma))$. The most likely value of $H(\sigma)$ is the value of E that maximizes $L_E \exp(-\beta E)$, where L_E counts the number of microstates that have energy E . Taking a log, we have $\log L_E - \beta E$, the competition between entropy and energy.

Bhaswar's comment on "Inference in Ising models":

The phase transitions relevant to statistics are not always the same as those that the physicists are interested in. We saw the BBP phase transition above, which can be interpreted as a phase transition in $M(\sigma) = \langle \sigma, \theta \rangle^2$ for a modified SK model.

Consider the problem of estimating β in the Ising model given a configuration σ of n sites. The log likelihood is $\ell(\sigma|\beta) = -\beta H_h(\sigma) - \log Z(\beta)$, so the MLE $\hat{\beta}$ satisfies $-H_h(\sigma) - \partial_\beta \log Z(\hat{\beta}) = 0$. We want asymptotics for $\hat{\beta}$, so we Taylor expand about the true β : $-H_h(\sigma) = \partial_\beta \log Z(\beta) + (\hat{\beta} - \beta) \partial_{\beta\beta} \log Z(\tilde{\beta})$. Rearranging, we find

$$\hat{\beta} - \beta = \frac{-H_h(\sigma) - \partial_\beta \log Z(\beta)}{\partial_{\beta\beta} \log Z(\beta)} + o(1) \quad (43)$$

The rate of convergence of $\hat{\beta} - \beta$ turns out to be controlled by the size of $\log Z(\beta)$. This log partition function undergoes a phase transition at $\beta = 1$, so that $\hat{\beta}$ is inconsistent for $\beta < 1$ and $\sqrt{n}$ -consistent for $\beta > 1$. (Actually, this should be the MPLE, fix...)

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